Marcos **Rodrigo**

AI Researcher & Machine Learning Engineer

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Summary

Ph.D. in Communication Technologies and Telecommunication Engineer with 6+ years developing deep-learning systems for computer vision, video understanding, and multimodal AI. Specialized in face recognition, sports video summarization, action recognition, and vision-language models, with end-to-end experience spanning dataset creation, model training, large-scale experimentation, evaluation, and GPU deployment. Published at CVPR Workshops, IEEE Access, Scientific Reports, and Multimedia Tools and Applications, with applied R&D collaborations involving Nokia and Airbus, plus university teaching and Master's-thesis supervision.

Skills: Python, PyTorch, TensorFlow/Keras, Hugging Face, Computer Vision, Video Understanding, Vision Transformers, Vision-Language Models, Multimodal Models, Large Language Models, Docker, Kubernetes, Linux, AWS, Git.

Work Experience

Universidad Politécnica de Madrid — *Madrid, Spain*

Post-graduate Researcher, Computer Vision Group — *2020 – 2026*

- Designed and evaluated deep-learning systems for computer vision and video understanding, including face recognition, video highlight detection, long-form sports-video summarization, and multimodal video analysis.
- Built end-to-end machine-learning pipelines covering data collection, dataset curation, preprocessing, model training, evaluation, ablation studies, and experimental reporting.
- Applied CNNs, Vision Transformers, 2D/3D CNNs, recurrent networks, Transformers, CLIP, and vision-language models to research problems in face analysis, action recognition, and text-guided video summarization.
- Delivered applied R&D in collaboration with Nokia and Airbus, translating research ideas into experimental prototypes, reproducible pipelines, and evaluated computer-vision solutions.
- Built and maintained a 17-node Kubernetes GPU cluster for teaching and research, including Docker images, JupyterHub profiles, NFS storage, NVIDIA GPU support, authentication, and HTTPS access.
- Supervised two Master's theses and supported teaching across three degree programmes in Telecommunication Engineering and Data Engineering.

Indra Sistemas S.A. — *Madrid, Spain*

Junior Programmer, Air Traffic Management — *2017 – 2018*

- Developed automation scripts for Voice-over-IP communication systems in air-traffic-management environments, supporting technical workflows in a high-reliability engineering context.
- Configured and tested telecommunications equipment, routers, switches, and network connections while collaborating with engineering teams under Scrum methodology.

DEYDE Calidad de Datos S.L. — *Madrid, Spain*

Data Coder & Trainee Programmer — *2016 – 2017*

- Processed, validated, and maintained structured databases of municipalities, streets, and postal-address records, supporting data-quality workflows for address-correction systems.
- Contributed to rule-based expert systems for postal-address correction and coding, bridging data operations, validation tasks, and software-support activities.

Education

Universidad Politécnica de Madrid — *Madrid, Spain*

Ph.D. in Communication Technologies and Systems — *2020 – 2025*

- Thesis: *Automatic Sports Video Summarization with Identity-Aware Highlight Selection*. Grade: *Cum Laude*.

Universidad Politécnica de Madrid — *Madrid, Spain*

M.Sc. in Telecommunication Engineering — *2018 – 2020*

- Thesis: *Automatic Highlight Detection in Martial Arts Tricking Videos*. Grade: 10/10.

Universidad de Alcalá de Henares — *Alcalá de Henares, Spain*

B.Sc. in Telecommunication Systems Engineering — *2013 – 2018*

- Thesis: *Acoustic Bird Classification Using MFCC Feature Extraction*. Grade: 9/10.

Selected Publications

ViMoCLIP: Augmenting Static CLIP Representations with Video-Motion Cues for Animal-Action Recognition
M. Rodrigo, E. Zhong, C. R. del-Blanco, C. Cuevas, F. Jaureguizar, N. García
IEEE/CVF CVPR Workshops (CV4Animals), 2025.

Text-Guided Sports Highlights: A CLIP-Based Framework for Automatic Video Summarization
M. Rodrigo, C. Cuevas, N. García
IEEE Access, 2025.

Comprehensive Comparison Between Vision Transformers and Convolutional Neural Networks for Face Recognition Tasks
M. Rodrigo, C. Cuevas, N. García
Scientific Reports 14, 21392, 2024.

Automatic Highlight Detection in Videos of Martial Arts Tricking
M. Rodrigo, C. Cuevas, D. Berjón, N. García
Multimedia Tools and Applications 83, 17109–17133, 2024.

UPM-GTI-Face: A Dataset for the Evaluation of the Impact of Distance and Masks in Face Detection and Recognition Systems
M. Rodrigo, E. González-Sosa, C. Cuevas, N. García
IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS), 2022.

Selected Engineering Projects

Olympus: Personal Multi-Agent AI Workspace *OpenClaw, Claude, FastAPI, SQLite, Telegram Bots*
- Built a 24/7 local multi-agent workspace coordinating specialized agents for planning, coding, task tracking, persistent memory, Telegram-based control, and GitHub-backed automation.

Job Finder: AI Job Monitoring Agent *LangGraph, Ollama, FAISS, SQLite, Streamlit*
- Developed a local-first agent that crawls public job sources, normalizes postings, and scores relevance through hybrid ranking with rules, semantic embeddings, and LLM-based fit assessment.

Paper Copilot: Research Paper Summarizer *LangChain, Ollama, Python, Streamlit*
- Built a local AI agent that reads research PDFs and generates structured, Notion-ready markdown summaries with metadata, methodology, key results, reference analysis, and extracted figures.

INDESIAhack: Adverse-Weather Detection for Ferrovial *CLIP, Azure, AWS SageMaker, ChatGPT API, MoE*
- Led a small team to build a model-of-experts pipeline combining CLIP, cloud AI services, and LLM-based reasoning to assess road visibility from traffic-camera images.

Generative AI Workflow Automation *Llama, Stable Diffusion, Python, Linux*
- Built a scheduled Linux service that combines Llama-based story generation with Stable Diffusion image generation to automate creative research-group workflows.

GPU Kubernetes Cluster for Teaching and Research *Kubernetes, Docker, JupyterHub, NFS, NVIDIA*
- Co-developed a 17-node GPU-enabled Kubernetes cluster with Docker profiles, JupyterHub, NFS storage, authentication, HTTPS access, and NVIDIA GPU support for teaching and research workloads.

Supplementary Education

- **How Diffusion Models Work**, DeepLearning.AI, 2025.
- **Generative AI with Large Language Models**, DeepLearning.AI, Coursera, AWS, 2024.
- **Deep Learning Specialization**, DeepLearning.AI, Coursera, 2023.
- **English as a Second Language**, University of California Los Angeles, 2015–2016.